

Section 1: Historical Overview of Psychopharmacology

This section covers the key milestones in psychopharmacology — a favourite for MCQ questions. Focus on who discovered what, when, and where.

Key Timeline — High Yield for MCQs

Year	Event	Key Person(s)
1915	Term 'psychopharmacology' coined — studied opioid alkaloids on rat behaviour in maze	Macht & Mora
1931	Antipsychotic properties of Rauwolfia serpentina reported (Reserpine)	Sen & Bose (Indian physicians, Calcutta)
1949	Lithium discovered for mania — initially used as lithium urate for uric acid solubility; later lithium ion itself found calming	Cade (Australia)
1950–52	Chlorpromazine shown to have antipsychotic effects (originally a presurgical antihistamine). Term 'neuroleptic' coined 1955	Delay & Deniker; Charpentier (France)
1952	First antidepressant — mood-lifting properties of Iproniazid (anti-TB drug) discovered	—
1954	First benzodiazepine — Chlordiazepoxide (Librium) discovered serendipitously	Leo Sternbach (Austrian)
1954	Reserpine rediscovered	Kline
1958	Haloperidol synthesised from pethidine (butyrophenone class)	Janssen
1958	Endogenous depression shown to respond best to Imipramine	Kuhn
1961	Second TCA — Amitriptyline introduced	—
1963	Cheese reaction mechanism for MAOI hypertension proposed	Blackwell
1970s	Fluoxetine initially tested as NA reuptake inhibitor — discarded. Later rediscovered as SSRI, marketed 1987	—
1988	Clozapine rediscovered via multicentre RCT vs chlorpromazine in treatment-resistant schizophrenia (4% vs 30% response)	Kane et al.
1989	Risperidone synthesised (atypical antipsychotic)	Janssen
1990s	'Prozac era' — widespread antidepressant prescribing; later black box warning for adolescents/children	
2005	CATIE study published — reappraisal of atypical vs typical antipsychotics. CuTLASS (UK) — no economic gain from atypicals	—

 **MCQ Traps — Remember These!**

- Psychopharmacology coined in 1915 — NOT 1950s
- Lithium discovered by Cade in AUSTRALIA — not Europe
- First antidepressant = Iproniazid (MAOI) — not imipramine
- Chlordiazepoxide (Librium) = first BZD, not diazepam
- Haloperidol synthesised from PETHIDINE — butyrophenone class
- Zimeldine was first SSRI synthesised by Carlsson — WITHDRAWN (hypersensitivity + demyelination)
- Fluoxetine originally tested as NA reuptake inhibitor — rediscovered as SSRI
- Kane 1988 clozapine trial: 30% response vs 4% chlorpromazine

Section 2: Classification of Psychotropics

Two ways to classify psychotropics: (1) Chemical structure and (2) Mechanism of action. Both are tested in MRCPsych Paper A.

2A. Classification by Chemical Structure — Antipsychotics

Chemical Class	Examples
Aliphatic phenothiazines	Chlorpromazine, Promazine, Triflupromazine
Piperidine derivatives	Thioridazine
Piperazine derivatives	Trifluoperazine, Fluphenazine, Perphenazine, Thioridazine
Butyrophenones	Haloperidol, Droperidol
Thioxanthenes	Thiothixene, Flupenthixol, Zuclopenthixol
Dihydroindoles	Molindone
Diphenylbutylpiperidine	Pimozide (long t _{1/2})
Dibenzoxapine	Loxapine
Benzisoxazole derivative	Risperidone
Substituted benzamides	Amisulpride, Sulpiride
Dibenzodiazepine	Clozapine
Dibenzothiazepine	Quetiapine
Thienobenzodiazepine	Olanzapine
Benzisothiazole	Ziprasidone
Arylpiperidylindole (quinolone)	Aripiprazole

2B. Classification by Chemical Structure — Antidepressants

Chemical Class	Examples	Notes
Tertiary amines (TCAs)	Imipramine, Amitriptyline, Clomipramine, Dosulepin, Trimipramine; also Venlafaxine	More sedating, more anticholinergic
Secondary amines (TCAs)	Desipramine, Amoxapine, Nortriptyline, Protriptyline; also Duloxetine	More potent, less sedating, more noradrenergic, less antihistaminic/anticholinergic than tertiary
Hydrazine derivatives (MAOIs)	Phenelzine, Isocarboxazid	Greater hepatotoxicity than Tranylcypromine (non-hydrazine)
Aminoketone	Bupropion	Amphetamine-like structure

2C. Classification by Chemical Structure — Other Psychotropics

Chemical Class	Drug
Azaspirodecanedione	Buspirone
Triazolopyridine	Trazodone, Nefazodone
Imidazopyridine	Zolpidem
Pyrazolopyrimidine	Zaleplon
Cyclopyrrolone	Zopiclone
Benzothiazolyl piperazine	Ziprasidone

2D. Classification by Mechanism of Action — Antidepressants

Mechanism	Abbreviation	Examples
Selective Serotonin Reuptake Inhibitor	SSRI	Citalopram, Paroxetine, Fluoxetine, Sertraline, Fluvoxamine, Escitalopram (S-enantiomer of citalopram)
Serotonin & Noradrenaline Reuptake Inhibitor	SNRI	Venlafaxine, Milnacipran, Duloxetine, Sibutramine
Noradrenaline Reuptake Inhibitor	NARI	Reboxetine
Noradrenaline Dopamine Disinhibitor	NDDI	Agomelatine
Noradrenergic & Specific Serotonergic Antagonist	NaSSA	Mirtazapine, Mianserin
Dopamine Reuptake Inhibitor	DARI	Bupropion
Reversible Inhibitor of Monoamine A Oxidase	RIMA	Moclobemide

Mechanism	Abbreviation	Examples
Serotonin Antagonist & Reuptake Inhibitor	SARI	Nefazodone, Trazodone

⚡ MCQ Traps — Classification

- Clozapine = DIBENZODIAZEPINE (not thienobenzodiazepine — that's olanzapine!)
- Quetiapine = DIBENZOTHIAZEPINE
- Olanzapine = THIENOBENZODIAZEPINE
- Pimozide has a notably long half-life
- Escitalopram = S-enantiomer of citalopram (SSRI)
- Agomelatine = NDDI (noradrenaline dopamine disinhibitor) — unique mechanism
- Moclobemide = RIMA — reversible and selective for MAO-A
- Tranylcypromine = non-hydrazine MAOI (less hepatotoxic than phenelzine/isocarboxazid)

2E. Novel Agents (Emerging — Worth Knowing)

- Xanomeline — M1/M4 muscarinic agonist; Phase 3 trials for schizophrenia; improves positive AND negative symptoms + cognition; high GI side effects
- Ketamine — NMDA receptor antagonist; rapid antidepressant effects in treatment-resistant depression
- Pomaglumetad methionil — metabotropic glutamate 2/3 receptor agonist; initial promise in schizophrenia but failed to replicate due to high placebo response; no affinity for dopamine receptors

Section 3: Principles of Rational Prescribing

Core Principles

Principle	Key Points
Watchful waiting	NICE recommends this BEFORE pharmacological intervention for depression and anxiety
Start low, go slow	Lowest possible dose for minimum duration; revise plan if no expected improvement
Therapeutic monitoring	Plasma monitoring useful for drugs with dose-dependent effects / narrow therapeutic window
Metabolic monitoring	Metabolic side-effects cause more days of life lost than most other adverse effects
Avoiding polypharmacy	Most guidelines explicitly recommend against combining antipsychotics
Informed consent	Discuss pros and cons before prescribing; patient makes informed decision

Principle	Key Points
Patient choice	NICE: patient must be supported to make final choice of specific psychotropic
Off-label use	Not illegal, but prescriber must inform patient and provide available evidence

Chemistry & Storage (MCQ Favourite!)

- Hygrophilicity — drugs gaining moisture from environment, reducing active excipients
- Deliquescence — extreme hygrophilicity: drug converts from crystalline to paste/liquid on contact with moist air
 - Classic example: SODIUM VALPROATE is deliquescent

Section 4: Placebo Effect

This is a very high-yield section with many specific facts that appear in MCQs. Learn the details carefully.

Key Definitions

Term	Definition
Placebo	Any intervention deliberately used for non-specific psychological or psychophysiological treatment effect
Placebo effect (research)	Any difference in outcome between placebo-treated group and untreated control group in unbiased experiment (Ernst 2001)
Active placebo	Has some inherent activity but NOT against the treated condition
Nocebo	A placebo that produces prominent side-effects. 'Nocebo effect' = negative consequences from placebo administration (Latin: 'I shall harm')
Placebo sag	Decrease in placebo effect with repeated/chronic administration
Efficacy paradox	Placebo effect may be disproportionately large for non-blinded therapies

Placebo coined by anaesthetist Beecher in 1955. Latin meaning: 'to please'.

Where Placebos Work Best

- Pain (especially severe pain — works BETTER for severe than mild pain)
- Disorders of autonomic sensation
- Neurohumoral conditions: nausea, blood pressure, bronchial asthma
- Psychiatric: depression (25–60%), anxiety, phobias, panic disorder (up to 70%)
- Mania: up to 25% placebo response
- Schizophrenia: 25–50% (criterion-dependent)

Where Placebos FAIL

- Hereditary degenerative disorders
- Toxic and metabolic syndromes
- Vascular events

Important Facts About Placebo Response

1. More chronically ill patients show LOWER placebo response rates
2. Placebo effects are seen with active drugs too — net effect of drug = pharmacological effect + placebo effect
3. Gender, suggestibility scores, and IQ do NOT affect placebo effect consistently
4. Three factors necessary for placebo action: nature of the disease + doctor-patient relationship + patient's past expectations/experience with treatment
5. No homogeneous placebo reactors exist — placebo response to one condition/doctor does NOT predict response to another
6. Placebo run-in approach (eliminating placebo reactors before trial) does NOT significantly lower placebo response rate or increase drug-placebo difference
7. Placebo response higher in trials with MORE than 2 arms (3-arm: 2/3 chance of active; 2-arm: 1/2 chance)
8. Placebos work better for CLINICAL than experimentally induced conditions
9. In depression and schizophrenia: little difference until ~2 weeks; full difference by 6 weeks; increases for several months

Pill Characteristics Affecting Placebo Response

- Capsules perceived stronger than tablets
- Larger pills have stronger placebo effect than smaller pills
- Multiple pills have stronger placebo effect than single pills
- Anxiety symptoms: respond better to GREEN tablets
- Depressive symptoms: respond better to YELLOW tablets (Schapira et al. 1970)
- Injections > oral medications for placebo effect
- Surgery likely strongest placebo effect of all

Theories Explaining Placebo Action

10. Natural remission theory — episodic disorders naturally remit; placebo use coincidental
11. Measurement regression — repeated measurement of a continuous variable moves towards population mean (regression to mean)
12. Conditioning theory — learned associations between medication vehicles (pills, injections = conditioned stimuli) and active medication (unconditioned stimulus) produce therapeutic conditioned responses
 - Classical conditioning: vehicles → conditioned stimuli for therapeutic effect
 - Operant conditioning: symptom improvement acts as internal reward; visceral learning
 - Placebo responses mediated by CONDITIONING for unconscious physiological functions (e.g. hormonal secretion)
 - Placebo responses mediated by EXPECTATION for conscious processes (e.g. pain, motor performance)

13. Endogenous opioids — placebo-induced analgesia partially reversed by NALOXONE; neuropeptide hypothesis holds for analgesia but NOT proven for depression
- Dopamine reward system increasingly implicated in placebo effects in psychotropic research
 - ACC (anterior cingulate cortex) activated during placebo analgesia — part of dopaminergic and opioid systems

Placebo vs Antidepressant Response in Depression

Feature	Placebo Response	Antidepressant Response
Onset	Abrupt, early in treatment	Gradual, occurs later
Persistence	Less likely to persist	More likely to persist
Neurobiology	Mayberg et al. 2002: PET scans nearly identical to fluoxetine responders	Similar neurobiological correlates — response may not be truly different

MCQ Traps — Placebo

- Placebo coined by Beecher (anaesthetist) 1955 — NOT a psychiatrist
- Nocebo = harmful placebo effect (Latin: I shall HARM). Placebo = I shall PLEASE
- Placebo SAG = decreasing effect with repeated administration
- Severe pain responds BETTER to placebo than mild pain (counterintuitive!)
- Mildly depressed respond BETTER to placebo than severely depressed (also counterintuitive!)
- Green = anxiety, Yellow = depression for tablet colour effect
- Naloxone partially reverses placebo analgesia — proves endogenous opioid involvement
- No consistent placebo reactor personality type exists

Section 5: Drug Approval Process

The 5 Phases of Drug Development

Phase	Name	Detail
Preclinical	Animal Studies	At least TWO different animal species required. Studies: mutagenicity, carcinogenicity, organ toxicity
Phase 1	Volunteers — Safety	Small group of healthy volunteers. Assesses safety, tolerability, pharmacokinetics. Usually open/uncontrolled studies
Phase 2	Patients — Effectiveness	Hundreds of patients with target disease vs placebo. Controlled trials / small RCTs
Phase 3	Patients — Comparative Efficacy	Extensive double-blind RCTs. Determines how well it works and common side effects. Superiority or equivalence vs standard treatment

Phase	Name	Detail
Phase 4	Post-Marketing Surveillance	After regulatory approval (FDA/MHRA). Detects LESS COMMON side effects that can lead to drug withdrawal

Drugs Withdrawn/Flagged After Post-Marketing Surveillance

Drug	Adverse Effect Detected
Nefazodone	Hepatotoxicity
Droperidol, Thioridazine	QT prolongation on ECG
Sertindole	Sudden cardiac death
Thalidomide (analgesic)	Phocomelia
Nomifensine	Hepatotoxicity
Zimeldine	Hypersensitivity reactions + Guillain-Barré syndrome
Remoxipride (sulpiride group)	Aplastic anaemia
Mianserin	Blood dyscrasias
MAOIs	Cheese reactions
Clozapine	Agranulocytosis

⚡ MCQ Traps — Drug Approval

- Phase 1 = SAFETY in HEALTHY VOLUNTEERS (not patients!)
- Phase 2 = effectiveness in patients vs placebo
- Phase 3 = comparative efficacy (double-blind RCT)
- Phase 4 = post-marketing surveillance — picks up RARE side effects
- Zimeldine = first SSRI — withdrawn due to Guillain-Barré + hypersensitivity
- Thioridazine AND Droperidol both cause QT prolongation (not just sertindole)
- Remoxipride is in the sulpiride (substituted benzamide) group — causes aplastic anaemia

Section 6: Medication Adherence

Key Definitions — Compliance vs Adherence vs Concordance

Term	Definition	Criticism
Compliance	Extent to which patient behaviour coincides with medical advice	Implies sole patient responsibility; criticised as paternalistic
Adherence	Includes patient choice; clinician and patient SHARE responsibility	Definitions usually dichotomous but adherence is rarely all-or-nothing

Term	Definition	Criticism
Concordance	Therapeutic alliance as a negotiation process; equal respect for both agendas	Most patient-centred model

Non-Adherence Rates — Learn These Numbers!

- Antipsychotics: 40–60%
- Mood stabilisers: 18–56%
- Antidepressants: 30–97% (median 63%)
- Non-adherent schizophrenia patients are 3.5 times more likely to relapse within 2 years

Measuring Adherence

Method	Details
Self-report	e.g. Tablet Routines Questionnaire (Scott and Pope 2002) — assesses daily routines and missed doses in past week/month
Pill counts	Adherence (%) = (pills taken ÷ pills prescribed) × 100 (Azrin and Teichner 1998)
Electronic methods	Devices attached to tablet bottle — record time and date each time bottle opened
Prescription monitoring	Frequency of dispensing used as proxy measure of adherence
Saliva/plasma/urine assay	Most objective measure; not available for all drugs; expensive, invasive; overestimates adherence for long half-life drugs

Factors Affecting Adherence

Factors with NO Influence on Adherence

- Age at illness onset
- Age at first hospitalisation
- Sex
- Socioeconomic status
- Marital status
- Ethnicity

Factors that REDUCE Adherence

- Lack of insight (most common cause)
- Asymptomatic stage of illness
- Cognitive deficits
- Comorbid alcohol and substance misuse
- Devaluation of medication by physician
- Fear of side-effects
- High frequency of daily doses
- Homelessness
- Long duration of illness (chronic diseases)

- Oral formulations — poorer adherence than depots
- Past history of non-adherence
- Polypharmacy
- Prophylactic or maintenance treatments
- Psychopathology: hostility, suspiciousness, disorganisation

Factors that INCREASE Adherence

- Presence of family support
- Liquid or sublingual formulations
- High enthusiasm from clinician
- Good patient-clinician relationship
- Continued access to clinicians

Health Belief Model of Adherence

Four categories a patient considers before making a medication decision:

14. Benefits
15. Costs
16. Susceptibility
17. Secondary benefits of medication and adherence

Interventions to Improve Adherence

Intervention	Mechanism
Psychoeducation alone	Primarily imparts knowledge WITHOUT attitudinal/behavioural change — largely INEFFECTIVE for adherence enhancement
Cognitive-based interventions	Targets patient's attitudes and beliefs towards medication; alters personal meaning of medication
Behaviour-modification interventions	Based on learning theory; uses reminders, self-monitoring, cues, reinforcements
Motivational interviewing	Patient expresses personal reasons for and against improving adherence
Compliance therapy	Brief intervention combining MI + cognitive approaches; explores ambivalence → consequences of stopping → analogies with chronic illness → pros/cons of medication

⚡ MCQ Traps — Adherence

- Concordance is the MOST patient-centred concept (not compliance)
- Most common cause of non-adherence = LACK OF INSIGHT
- Ethnicity, sex, SES, marital status have NO effect on adherence
- Psychoeducation alone is largely INEFFECTIVE for adherence
- Depot > oral formulations for adherence
- Non-adherent schizophrenia patients: 3.5x more likely to relapse at 2 years
- Saliva/plasma assay = most OBJECTIVE but overestimates adherence for long half-life drugs

Quick Reference: Key Names & Numbers

What	Who / When / Number
'Psychopharmacology' coined	Macht & Mora, 1915
Rauwolfia antipsychotic properties	Sen & Bose (Calcutta), 1931
Lithium for mania	Cade (Australia), 1949
Chlorpromazine antipsychotic	Delay & Deniker / Charpentier (France), 1950–52
'Neuroleptic' coined	Delay, 1955
First antidepressant (Iproniazid)	1952
First BZD (Chlordiazepoxide/Librium)	Leo Sternbach (Austrian), 1954
Reserpine rediscovered	Kline, 1954
Haloperidol synthesised	Janssen, 1958 (from pethidine)
Imipramine for endogenous depression	Kuhn, 1958
Cheese reaction mechanism (MAOIs)	Blackwell, 1963
Clozapine RCT rediscovery	Kane et al., 1988 (30% vs 4%)
Risperidone synthesised	Janssen, 1989
CATIE study	2005
Placebo coined	Beecher (anaesthetist), 1955
Placebo response in depression	25–60%
Placebo response in panic disorder	Up to 70%
Non-adherence in antipsychotics	40–60%
Non-adherence in antidepressants	30–97% (median 63%)
Relapse risk with non-adherence (schizophrenia)	3.5x within 2 years
Pill count adherence formula	$(\text{Pills taken} \div \text{pills prescribed}) \times 100$ (Azrin & Teichner 1998)
Tablet Routines Questionnaire	Scott and Pope, 2002